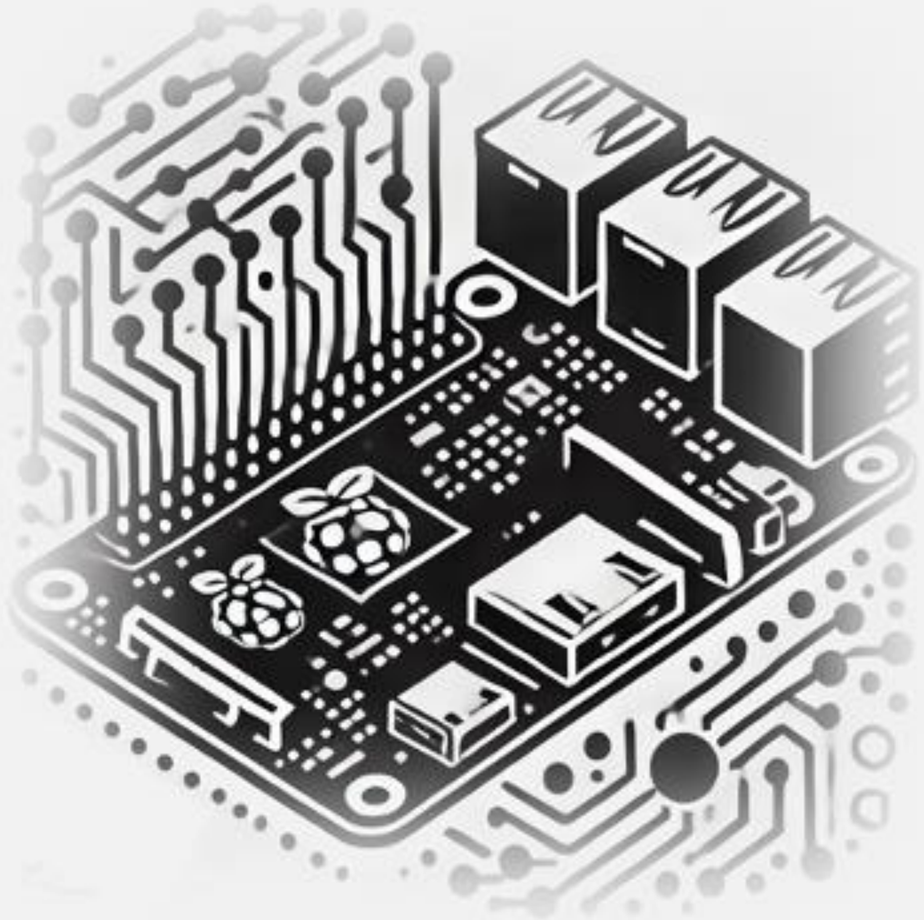
IESTI05 – Edge AI

Machine Learning System Engineering

11. Object Detection with YOLO: Fundamentals

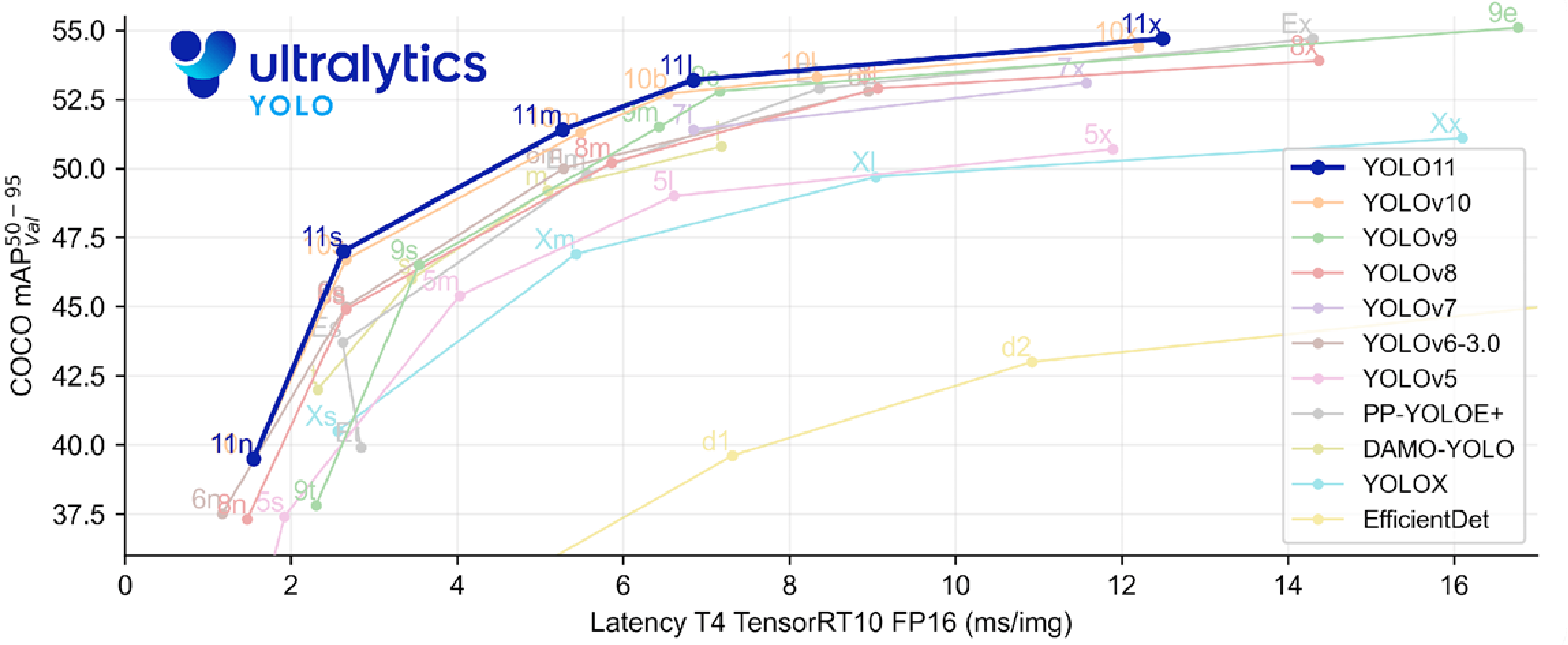


UNIFEI

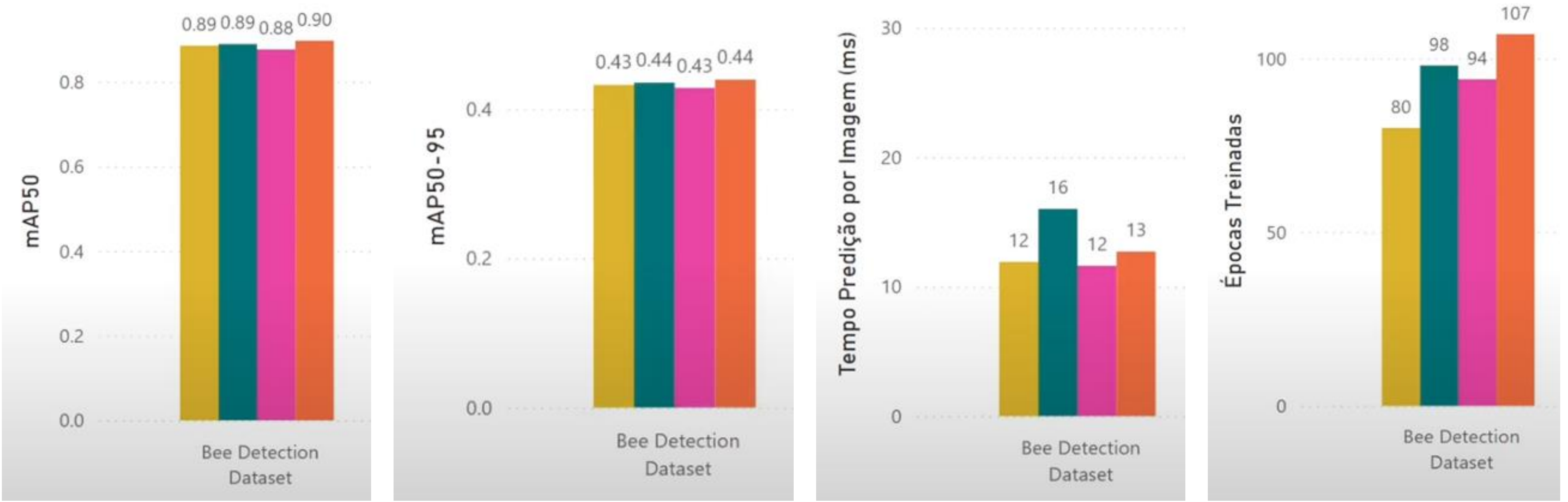
Marcelo Rovai
Professor Honoris Causa

YOLO
Fundamentals

Ultralytics YOLO11



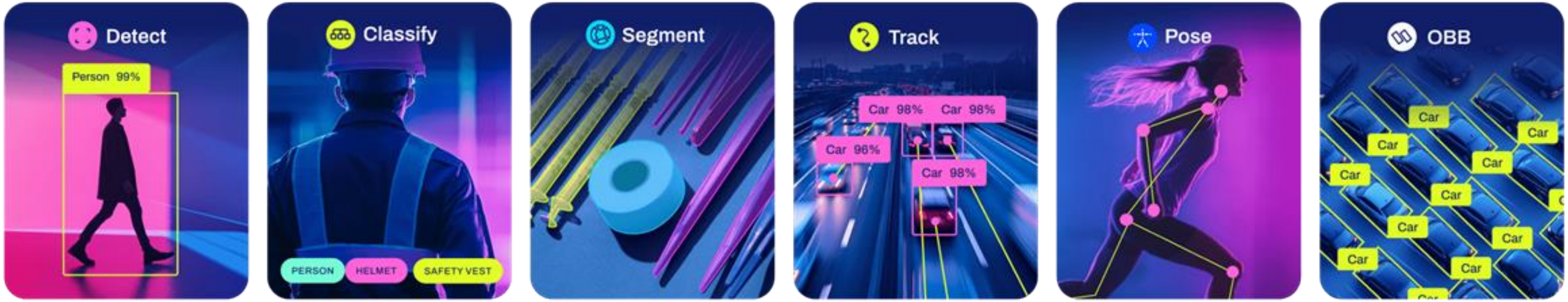
YOLO Versions Comparison



[Spreadsheet]

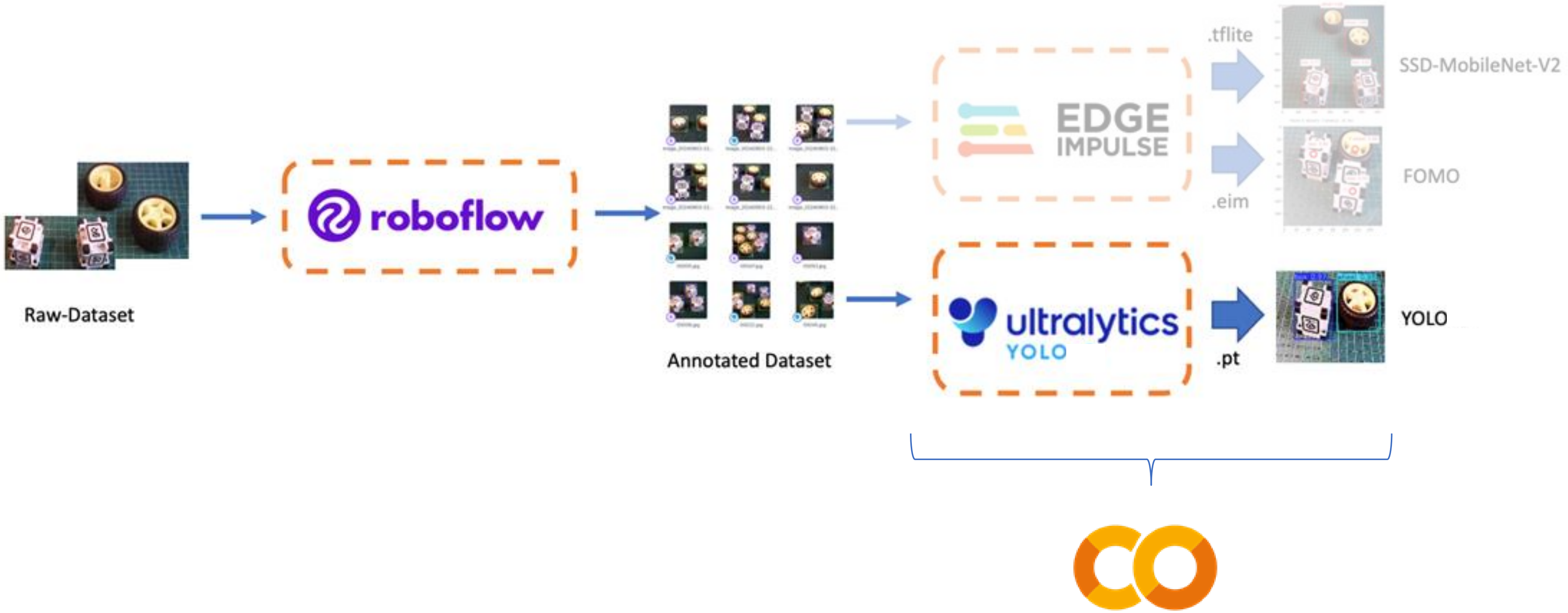
https://youtu.be/F8m0qo_gMbc?si=ADW8u6Q_yAM4GlY4

Ultralytics YOLO11

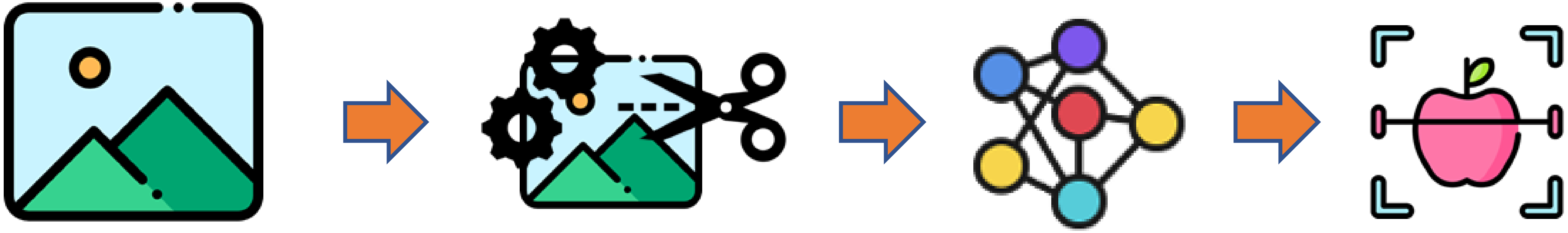


Oriented Bounding Boxes

YOLO with Roboflow and Ultralytics



Object Detection: Inference Pipeline



Input Image

Pre-Process

Detection Model

Predict Labels & Boulding Boxes

YOLOv11 - Ultralytics

Setting up a Virtual Environment

1. Create a virtual environment with access to system packages to manage dependencies:

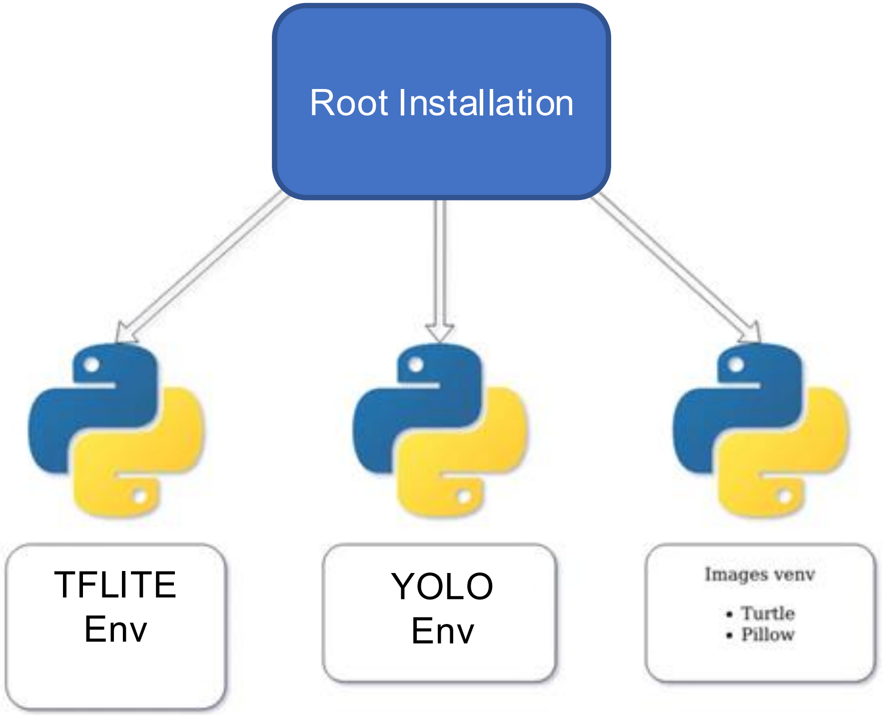
```
python3 -m venv ~/yolo --system-site-packages
```

2. Activate the environment:

```
source ~/yolo/bin/activate
```

To exit the virtual environment, use:

```
deactivate
```



Creating a working directory

marcelo_rovai — mjrovai@raspi-zero: ~/Documents/YOLO — ssh mjrovai@192.168.4.210 — 64×10

```
(yolo) mjrovai@raspi-zero:~ $ cd Documents/
(yolo) mjrovai@raspi-zero:~/Documents $ mkdir YOLO
(yolo) mjrovai@raspi-zero:~/Documents $ cd YOLO
(yolo) mjrovai@raspi-zero:~/Documents/YOLO $ mkdir models
(yolo) mjrovai@raspi-zero:~/Documents/YOLO $ mkdir images
(yolo) mjrovai@raspi-zero:~/Documents/YOLO $ ls
images  models
(yolo) mjrovai@raspi-zero:~/Documents/YOLO $
```

Install the Ultralytics packages

- Update the packages list, install pip, and upgrade to the latest:
  - `sudo apt update`
  - `sudo apt install python3-pip -y`
  - `pip install -U pip`
- Install the PyTorch and Ultralytics packages:
  - `mkdir -p ~/tmp`
  - `export TMPDIR=$HOME/tmp`
  - `pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu`
  - `pip install ultralytics`
  - `sudo reboot`

Note: Installing Ultralytics packages takes several minutes

YOLO11n versus v8n - Inference

```
marcelo_rovai — mjrovai@raspi-zero: ~/Documents/YOLO — ssh mjrovai@192.168.4.210 — 107×17
(yolo) mjrovai@raspi-zero:~/Documents/YOLO $ yolo predict model=yolo11n.pt source=bus.jpg
Ultralytics 8.3.191 🚀 Python-3.11.2 torch-2.8.0+cpu CPU (Cortex-A53)
YOLO11n summary (fused): 100 layers, 2,616,248 parameters, 0 gradients, 6.5 GFLOPs

image 1/1 /home/mjrovai/Documents/YOLO/bus.jpg: 640x480 4 persons, 1 bus, 3222.3ms
Speed: 186.6ms preprocess, 3222.3ms inference, 148.3ms postprocess per image at shape (1, 3, 640, 480)
Results saved to runs/detect/predict4
💡 Learn more at https://docs.ultralytics.com/modes/predict
(yolo) mjrovai@raspi-zero:~/Documents/YOLO $ yolo predict model='yolov8n' source='bus.jpg'
Ultralytics 8.3.191 🚀 Python-3.11.2 torch-2.8.0+cpu CPU (Cortex-A53)
YOLOv8n summary (fused): 72 layers, 3,151,904 parameters, 0 gradients, 8.7 GFLOPs

image 1/1 /home/mjrovai/Documents/YOLO/bus.jpg: 640x480 4 persons, 1 bus, 1 stop sign, 6870.6ms
Speed: 190.2ms preprocess, 6870.6ms inference, 157.0ms postprocess per image at shape (1, 3, 640, 480)
Results saved to runs/detect/predict5
💡 Learn more at https://docs.ultralytics.com/modes/predict
(yolo) mjrovai@raspi-zero:~/Documents/YOLO $
```

Export Models to NCNN format

```
yolo export model=yolov8n.pt format=ncnn
yolo export model=yolo11n.pt format=ncnn
```

marcelo_rovai — mjrovai@raspi-zero: ~/Documents/YOLO — ssh mjrovai@192.168.4.210 — 135×9

```
(yolo) mjrovai@raspi-zero:~/Documents/YOLO $ yolo predict task=detect model='./models/yolov8n_ncnn_model' source='./images/bus.jpg'
Ultralytics 8.3.191 🚀 Python-3.11.2 torch-2.8.0+cpu CPU (Cortex-A53)
Loading ./models/yolov8n_ncnn_model for NCNN inference...

image 1/1 /home/mjrovai/Documents/YOLO/images/bus.jpg: 640x640 4 persons, 1 bus, 3566.8ms
Speed: 484.4ms preprocess, 3566.8ms inference, 1519.0ms postprocess per image at shape (1, 3, 640, 640)
Results saved to runs/detect/predict10
💡 Learn more at https://docs.ultralytics.com/modes/predict
(yolo) mjrovai@raspi-zero:~/Documents/YOLO $
```

marcelo_rovai — mjrovai@raspi-zero: ~/Documents/YOLO — ssh mjrovai@192.168.4.210 — 138×9

```
(yolo) mjrovai@raspi-zero:~/Documents/YOLO $ yolo predict task=detect model=models/yolo11n_ncnn_model imgsz=640 source=./images/bus.jpg
Ultralytics 8.3.191 🚀 Python-3.11.2 torch-2.8.0+cpu CPU (Cortex-A53)
Loading models/yolo11n_ncnn_model for NCNN inference...

image 1/1 /home/mjrovai/Documents/YOLO/images/bus.jpg: 640x640 4 persons, 1 bus, 1034.6ms
Speed: 1764.9ms preprocess, 1034.6ms inference, 3406.3ms postprocess per image at shape (1, 3, 640, 640)
Results saved to runs/detect/predict11
💡 Learn more at https://docs.ultralytics.com/modes/predict
(yolo) mjrovai@raspi-zero:~/Documents/YOLO $
```

YOLO26n (and ONNX) - Inference

marcelo_rovai — mjrovai@raspi-zero2w-26: ~/Documents/YOLO — ssh mjrovai@192.168.4.210 — 105×9

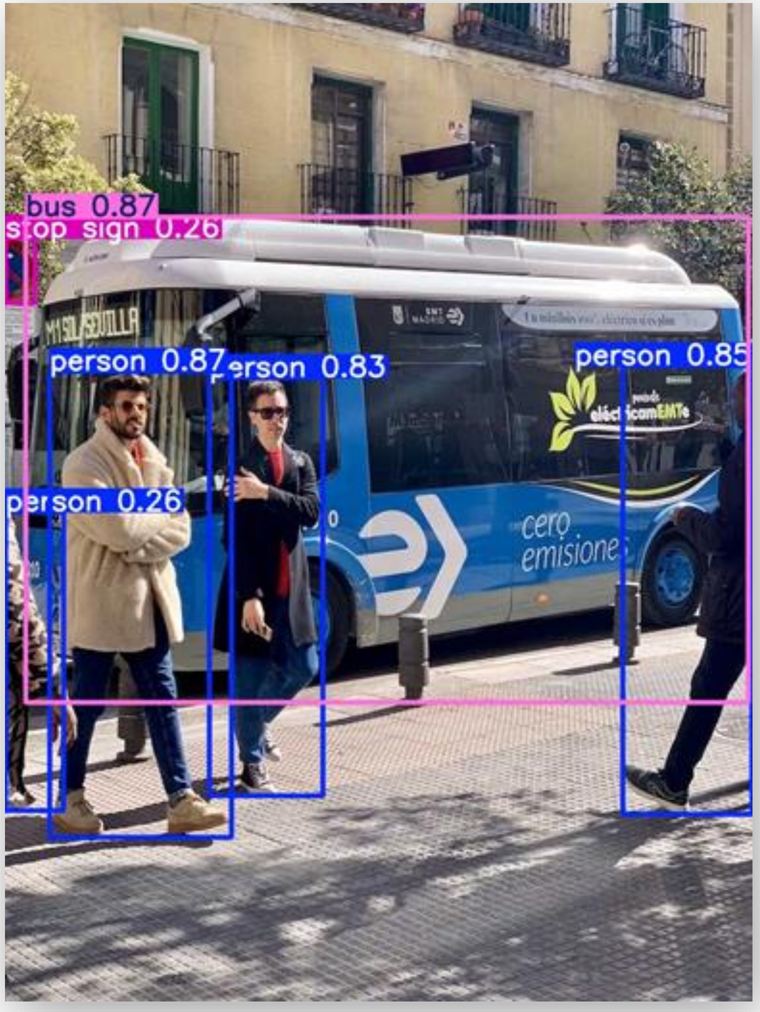
```
(yolo) mjrovai@raspi-zero2w-26:~/Documents/YOLO $ yolo predict model=yolo26n.pt source=bus.jpg
Ultralytics 8.4.121 🚀 Python-3.13.5 torch-2.13.0+cpu CPU (aarch64)
YOLO26n summary (fused): 122 layers, 2,408,932 parameters, 0 gradients, 5.5 GFLOPs

image 1/1 /home/mjrovai/Documents/YOLO/bus.jpg: 640x480 4 persons, 1 bus, 2750.3ms
Speed: 92.5ms preprocess, 2750.3ms inference, 35.9ms postprocess per image at shape (1, 3, 640, 480)
Results saved to /home/mjrovai/Documents/YOLO/runs/detect/predict-4
💡 Learn more at https://docs.ultralytics.com/modes/predict
(yolo) mjrovai@raspi-zero2w-26:~/Documents/YOLO $
```

marcelo_rovai — mjrovai@raspi-zero2w-26: ~/Documents/YOLO — ssh mjrovai@192.168.4.210 — 107×10

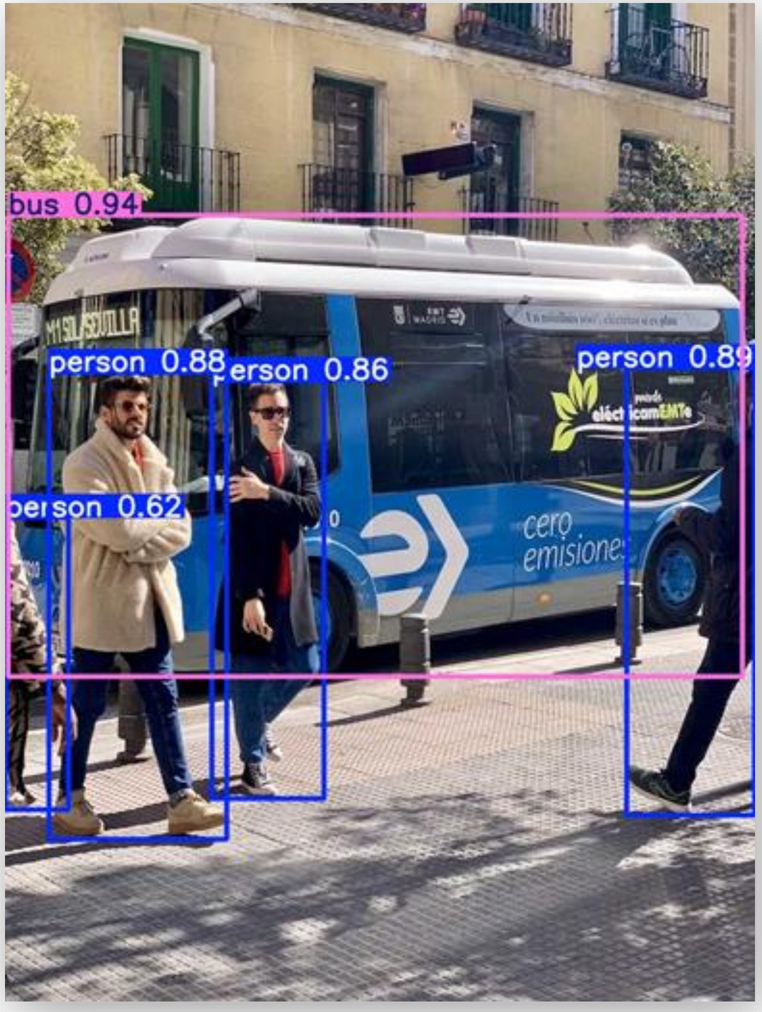
```
(yolo) mjrovai@raspi-zero2w-26:~/Documents/YOLO $ yolo predict task=detect model=./models/yolo26n.onnx sour
ce=./images/bus.jpg imgsz=640
Ultralytics 8.4.121 🚀 Python-3.13.5 torch-2.13.0+cpu CPU (aarch64)
Loading ./models/yolo26n.onnx for ONNX Runtime inference...
Using ONNX Runtime 1.29.0 with CPUExecutionProvider

image 1/1 /home/mjrovai/Documents/YOLO/images/bus.jpg: 640x640 4 persons, 1 bus, 2274.2ms
Speed: 305.3ms preprocess, 2274.2ms inference, 264.8ms postprocess per image at shape (1, 3, 640, 640)
Results saved to /home/mjrovai/Documents/YOLO/runs/detect/predict-13
💡 Learn more at https://docs.ultralytics.com/modes/predict
```

YOLOv11 versus v8 - Inference

YOLOv8



YOLOv11



Other Computer Vision Tasks

YOLO Segmentation Result

YOLO Pose Estimation Result

Replacing zram with a swap file on the SD card

Create a 1 GB swap file on the SD card

```
sudo fallocate -l 1G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
echo '/swapfile none swap sw,pri=-2 0 0' | sudo tee -a /etc/fstab
```

Disable zram permanently

```
sudo systemctl mask systemd-zram-setup@zram0.service
sudo reboot
```

Confirm: /swapfile should be the only entry

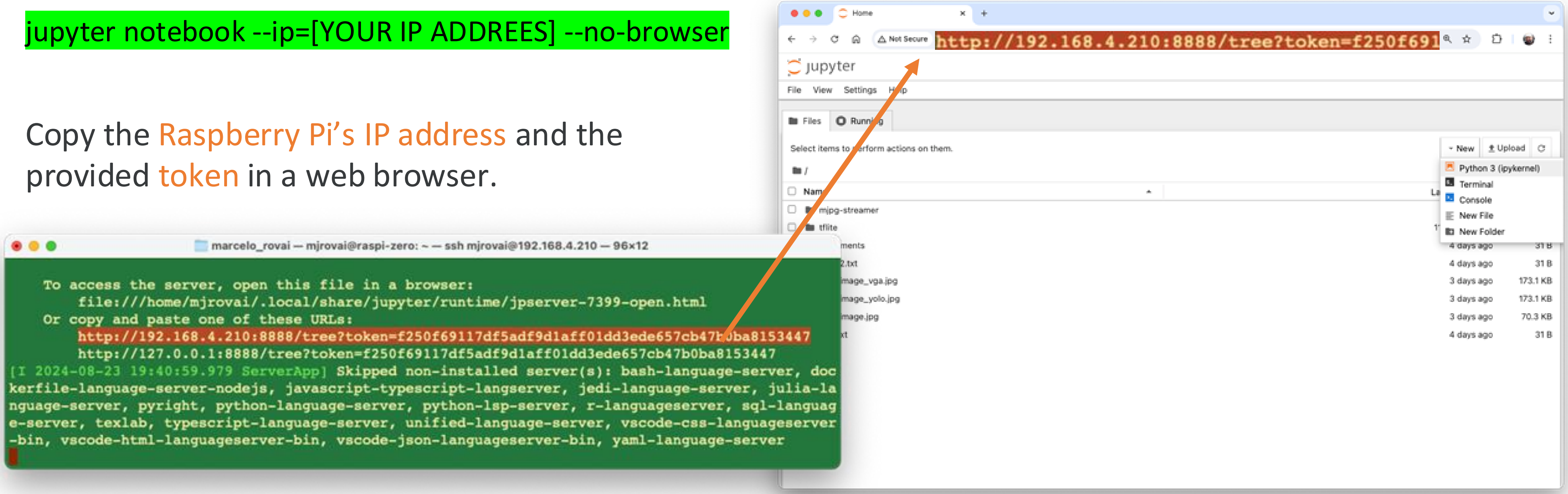
```
swapon --show
free -h
```

Setting up Jupyter Notebook

```
pip install jupyter jupyterlab notebook
```

```
jupyter notebook --ip=[YOUR IP ADDREES] --no-browser
```

Copy the Raspberry Pi’s IP address and the provided token in a web browser.



Raspberry Pi Inference:

YOLO_Model_Predictions_with_Ultralytics.ipynb

Questions?

Prof. Marcelo J. Rovai

rovai@unifei.edu.br

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